Hardening - Standard Installation

Step-by-step guide to protect the Raspberry Pi SD card (no boot settings are changed)

Read this first

- Follow only the steps in this document. **Do not edit cmdline.txt** and do not change any other boot setting.
- The overlay (read-only) mode must be **off** before you start.
- If you can, make an image backup of the SD card first. Test on one unit before installing on others.

1. What this does

The broker writes config.json, status.json, rfid1.json, rfid2.json and conf.d/bridge.conf on every scan or setting change. If power is cut during a write, a file can be left empty or half-written, and continuous log writing makes SD card damage more likely. This package fixes both problems **without changing any existing code**:

- **Atomic writes** - each protected file is written to a temporary file, flushed to the card, then renamed over the real file. After a power cut the file holds either the complete old data or the complete new data, never a half-written file.
- **Less writing to the SD card** - the application logs/ folder, the system journal and swap move to RAM; noatime is enabled; Mosquitto stops persisting and logging to files. The logger can stay enabled without writing to the card (logs are lost at reboot).

2. What is in the package

File	Purpose
hardening/sitecustomize.py	Atomic-write layer. Loaded automatically by Python; falls back to normal behaviour on any error.
harden_sd.sh	Installs the systemd drop-ins for send.service and ipservice.service, edits fstab, moves logs and journal to RAM, disables swap, adjusts Mosquitto. Safe to run more than once. Refuses to run while the overlay is active.

Both are new files. No existing file is changed. Assumptions: user **alice**, folder **/home/alice/broker** (change HOME_BASE at the top of harden_sd.sh if different). Backups are created automatically: fstab.bak and mosquitto.conf.bak.

3. Installation

Step 1 - Check that the overlay is off

```
findmnt -no FSTYPE /
```

It must print **ext4**. If it prints **overlay**, disable the overlay first: `sudo raspi-config nonint do_overlayfs 1` and reboot.

Step 2 - Copy the files

Copy the **hardening/** folder and **harden_sd.sh** (from the 1_COPY_TO_BROKER folder of the package) into **/home/alice/broker/**. No existing file is overwritten.

Step 3 - Run the script

```
cd /home/alice/broker
chmod +x harden_sd.sh
sudo ./harden_sd.sh
```

Step 4 - Reboot

```
sudo reboot
```

4. Verify after the reboot

```
findmnt /home/alice/broker/logs     # must show: tmpfs
sudo systemctl show send -p Environment | grep PYTHONPATH
systemctl status send                # must be active (running)
```

Only when **logs/ is tmpfs**, you may re-enable the logger by setting `"logger": "enable"` in config.json. If logs/ is not tmpfs, keep the logger disabled - otherwise it writes to the SD card again.

5. Testing

Change a setting from the web page (or send a scan), then cut power at different moments: during a scan, while saving a setting, during the automatic reboot. Repeat 10-20 times. After each boot confirm the newest setting is present and each file is valid JSON:

```
python3 -m json.tool /home/alice/broker/config.json
python3 -m json.tool /home/alice/broker/rfid2.json
```

At most the single update in progress at the moment of the power cut may be lost. The file itself must always be valid.

6. If something goes wrong (undo)

Atomic-write layer (works over SSH):

```
sudo rm /etc/systemd/system/send.service.d/10-hardening.conf
sudo rm /etc/systemd/system/ipservice.service.d/10-hardening.conf
sudo systemctl daemon-reload && sudo systemctl restart send ipservice
```

RAM logs / fstab: restore the backup, then reboot:

```
sudo cp /etc/fstab.bak /etc/fstab
sudo reboot
```

Mosquitto: `sudo cp /etc/mosquitto/mosquitto.conf.bak /etc/mosquitto/mosquitto.conf && sudo systemctl restart mosquitto`

7. Risks and limits

Item	Risk	Notes
sitecustomize.py	Low	Fails soft: errors fall back to normal file writes. Tested with the file-writing patterns used by the broker; not yet tested on a Pi with the full application.
fstab tmpfs for logs	Low	Marked nofail so a failed mount does not block boot. A wrong user ID prevents logging; logs are lost at reboot.
Swap off, journal in RAM	Low	Only matters on very small RAM (logs tmpfs uses up to 100 MB).
Mosquitto persistence off	Low	Queued messages do not survive a broker restart.

What software cannot guarantee. A power cut while the SD card's own firmware is writing can damage consumer-grade cards regardless of the operating system. The strongest additional protections are an industrial or pSLC-class SD card and a small UPS or supercapacitor module that triggers a clean shutdown.